

V-NET-VGG16: Hybrid deep learning architecture for optimal segmentation and classification of multi-differentiated liver tumors

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ABSTRACT

Liver cancer is a leading cause of cancer-related mortality worldwide, underscoring the importance of early and accurate diagnosis. This study aims to develop an automatic system for liver tumor detection and classification using Computed Tomography (CT) images, addressing the critical challenge of accurately segmenting liver tumors and classifying them as benign, malignant, or normal tissues.

The proposed method combines two advanced deep learning models: V-Net for tumor segmentation and VGG16 for classification. A liver CT dataset augmented with various transformations, was used to enhance the model's robustness. The data was split into training (70 %) and testing (30 %) sets. The V-Net model performs the segmentation, isolating the liver and tumor regions from the CT images, while VGG16 is used for the classification of tumor types based on the segmented data.

The results demonstrate the effectiveness of this hybrid approach. The V-Net model achieved a Dice score of 97.34 % for accurate tumor segmentation, while the VGG16 model attained a classification accuracy of 96.52 % in differentiating between benign, malignant, and normal cases. These results surpass several existing state-of-the-art approaches in liver tumors analysis, demonstrating the potential of the proposed method for reliable and efficient medical image processing.

In conclusion, the hybrid V-Net and VGG16 architecture offers a powerful tool for the segmentation and classification of liver tumors, providing a significant improvement over manual segmentation methods that are prone to human error. This approach could aid clinicians in early diagnosis and treatment planning. Future work will focus on expanding the dataset and applying the method to other types of cancer to assess the model's generalizability and effectiveness in broader clinical settings.

1. Introduction

In recent decades, significant progress has been made in medical image acquisition techniques, including magnetic resonance imaging (MRI), computed tomography (CT), ultrasound, positron emission tomography (PET), and X-rays. These advancements have greatly enhanced the capabilities for detecting and diagnosing liver diseases [1]. The manual evaluation of a 3D volume in a slice-by-slice method is used in traditional approaches to segment the liver and tumor from CT or MRI scans. Although manual segmentation techniques are used in many traditional computer-aided diagnostic (CAD) systems, they are time-consuming and rely on human interpretations and visual

observation. Also, there is a possibility that some slices of 3D scanned images will be misinterpreted as a disease or tumors. As a result, hundreds of new CAD systems have lately been developed to make the diagnosis system more efficient and error-free. The significance of developing an effective CAD system for segmenting, classifying, and diagnosing liver and liver tumors from 3D volumetric images can be illustrated by the fact that liver cancer is one of the leading causes of death worldwide [2,3]. Liver segmentation provides information on the anatomical features of the liver which is being used to plan interventions such as tumor removal, living donor transplants, and minimally invasive surgery using computer-assisted surgery [4].

Manual segmentation of the liver is based on the careful intuition and

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investigation of an operator, which is time-consuming and may cause human error. From the last few years, different machine learning algorithms are used to develop automatic liver and liver lesion extraction methods. However, its accurate segmentation is still an open challenge. Recently, deep learning has gained great attention in terms of segmentation in biomedical images acquired using different modalities [5].

To explore the potential of computer-aided analysis, a useful image processing techniques can be employed for liver tissue classification in order to assist the expert in diagnosis and decision-making method [6].

After the successful implementation of the U-Net (encoder-decoder) model by Falk et al. [7], deep learning techniques for medical image segmentation have become significantly popular. AI-based CAD systems are gaining popularity due to the availability of high-performance computing and hardware resources for the implementation of deep learning algorithms. These CAD systems [8] use a model that has already been trained, as opposed to manual CAD systems. During the training phase, these AI models are trained on data from a specific domain, such as abdominal images, which includes the liver, and then acquire features of the liver, such as shape, texture, and intensity values. These AI-based CAD systems can be used to interpret and manipulate illnesses or tumors without the need for human intervention once the model is trained, saving time and effort for medical doctors in the disease diagnosis. The goal of this paper is to review state-of-the-art deep learning based approaches for segmenting liver and liver tumors from CT and MRI volumetric images [9,10].

In the present study, we use the combined V-Net and VGG16 architecture in order to enhance the segmentation and classification of liver tumors. The proposed strategy is separated into three chief stages.

- (1) Data collection: CT images have been incorporated into the construction of the database. Expanding the dataset can significantly enhance the performance of the proposed segmentation and classification process.
- (2) CT image segmentation: The proposed approach consists of a pertinent segmentation of CT dataset. First, the tumor region is extracted using a V-Net algorithm. This step contributes to the accuracy and success of the proposed analysis process.
- (3) Classification: By the use of the significant VGG16 method, a rapid classification is achieved from the tumors structure and to differentiate between Benign, Malignant and Normal.

This paper is structured into six main sections. Section 2 provides a survey of articles utilizing deep learning approaches for liver and liver tumor segmentation. Section 3 delves into the materials and deep learning architectures employed for medical image segmentation. In Section 4, the results of liver and liver tumor segmentation and classification are presented. Section 5 outlines the evaluation metrics utilized for CT image segmentation and classification. Finally, Section 6 offers concluding remarks on the study.

2. Related work

Artificial intelligence methods [11] have significance in several research applications for optimal characterization of the liver tumor. Regarding liver cancer detection, numerous systems are realized containing machine learning methods, watershed transform and region-based methods. A computer aided diagnosis (CAD) method proposed by Huang et al. [12] for segmenting and identifying the liver tumors using CT images. 81.7 % of tumor classification precision is achieved in the extension work based on the auto-covariance texture features [13]. In clinical diagnosis of hepatocellular carcinoma, Priya et al. [14] projected an important computational method focused on a particle swarm optimization technique. Newly, deep learning; in which neural networks and images are respectively processed of numerous layers and multiple types of filters; has been drawing attention as regards effective pattern recognition [15]. Deep convolutional neural

network (CNN) method permits all of the information limited in the input image to be employed depending on the feature set picked out in convolutional machine learning. These methods [15,16] achieve a pertinent performance in the pattern recognition of CT images which can successfully distinguish liver masses without radiologist intervention. Deep learning techniques have achieved remarkable success across various computer vision domains, including region detection [17], semantic segmentation [18], and medical image classification [19]. This allows obtaining important progress in automated medical image diagnosis [20]. Melendez et al. [21] carried out many-instance learning focused on chest X-rays in order to detect the tuberculosis. It attained an AUC of 0.86. In the work of Huo et al. [22], a liver segmentation is appreciated based on CNN and globally optimized surface evolution, having 97 % of mean Dice similarity coefficient (DSC). To classify skin cancer, Esteva et al. [23] trained a CNN and completed advanced accuracy compared to dermatologists. Besides, CNN has been applied in the segmentation process of different substructures for ventricles [24], and liver [25]. CNN remains one of the most effective deep learning methods employed in various applications. Elshaer et al. [26] successfully reduced computation time by employing two trained deep CNN models: the first for liver region extraction and the second for minimizing blurriness resulting from image re-sampling and detecting small lesions that might have been missed. Similarly, Sharma et al. [27] used a CNN system operating on image patches, with each patch centered on a pixel of interest. The proposed model has outperformed pixel-based neural network, pixel-based CNN and patch-based neural network by 24.36 %, 24.23 % and 11.52 %, respectively, in overall classification accuracy. By combining the proposed deep CNN and the huge collection of medium-resolution remote sensing data, we believe that much more accurate land cover datasets can be produced over large areas.

According to Moghbel et al. [28], liver segmentation has nearly reached human-like precision, however, robust liver tumor segmentation remains a significant challenge. One of the reasons is the non-availability of public datasets with liver and liver tumor labels, and access to private datasets is very limited because of privacy issues and the tedious process of annotating the images. Therefore, only publicly available datasets are mostly used to develop new and computer-aided diagnosis (CAD) systems. It is evident from the past public challenges for liver and tumor segmentation tasks that most of the methods published were interactive and semi-automatic approaches. Recently, automatic methods for liver and liver tumor segmentation based on deep learning have been developed. In 2019, the CHAOS challenge was held to analyze the capability of multi-modal systems that utilized voxel information for liver segmentation from both CT and MRI volumetric images [29]. These public challenges are crucial not only in making much-needed datasets available, but also contributing in the development of robust CAD systems and improving healthcare system efficiency. As a result, these challenges serve as a benchmark for comparing state-of-the-art methods for liver and tumor segmentation.

This study makes an effort to provide a detailed overview of all published deep learning algorithms for liver and liver tumor segmentation, as well as the results of these architectures in terms of various assessment metrics. The review of public challenges and competitions to evaluate the performance of various methods for segmenting the liver and hepatic tumors can be used to visualize and compare different state-of-the-art approaches, assisting researchers in reproducing results from these methods, developing novel methodologies, or application of algorithms for specific datasets. The publicly accessible datasets for liver and tumor segmentation, as well as their modalities, format, size, and availability of labels for liver and tumor, have been thoroughly discussed. The relevant discussions and insights shed light on the problem of liver and tumor segmentation, as well as the modality selection, data, and design of computer-assisted diagnosis systems [29].

Different approaches used CNN for different medical image segmentation applications, where the U-net and V-Net models present two important deep learning methods that have simpler models recognized

as convolutional networks [30]. This unsupervised learning offers a multilevel structure layer by-layer, automatically selecting increasingly more representations from the layers. The CNN Encoder-Decoder architecture mostly averts the gradient problem that can occur when training a standard neural network (without pre-initialization). U-net pre-training increases the performance of the model by optimizing the CNN architecture and avoiding overfitting compared to others learning methods [31].

The work presented in Ref. [32] addresses the setup and results of the Liver Tumor Segmentation Benchmark (LiTS), held in conjunction with biomedical imaging conferences in 2017 and 2018. The dataset, created with seven hospitals, contains diverse primary and secondary liver tumors. Seventy-five algorithms were trained on 131 CT volumes and tested on 70 new images. No single algorithm performed best for both liver and tumor segmentation. The highest Dice score for liver segmentation was 0.963, while tumor segmentation scores ranged from 0.674 to 0.739 across the events. Further analysis revealed that top segmentation algorithms didn't perform as well for tumor detection, with the best detection method achieving recalls from 0.458 to 0.554, highlighting the need for continued research.

Due to the deep learning methods need an important data base, these two architectures can supply accurate models even with moderately small datasets. An enhanced fully convolutional neural network (CNN) concept is obtained by the U-Net method [33] via CNN layers followed by up-sampling ones. In this paper, an automated segmentation is established using CT liver cancer images. The included lesions are in low-contrast heterogeneous medical volumes. The proposed contribution is divided into two folds (segmentation and classification of CT images). A fully convolutional CNN using V-Net method is firstly trained in order to prove the adaptability to complicated liver lesions detection. To automatically identify benign and malignant tumors of the liver using CT images, this paper reports these outcomes and elaborates a CAD scheme. The proposed process is applied in order to compare the extracted data by means of their ability for differentiating between malignant and benign cases. Performance analysis process is completed through means of accuracy and other measures using several experimentations.

In the work of Anter et al. [34], authors propose a novel dynamic approach that enhances the fuzzy c-means (FCM) clustering algorithm, specifically focusing on improving accuracy, speed, and optimal convergence. Their approach, named CALOFCM, integrates fast-FCM, chaos theory, and a bio-inspired ant lion optimizer (ALO) to optimize the segmentation process. Validated on abdominal liver CT images, the method demonstrates superior detection and segmentation performance compared to other techniques, aiding earlier and more systematic clinical diagnosis of hepatic lesions.

The study presented in Ref. [35] developed an enhanced approach for segmenting liver tumors in abdominal CT scans using neutrosophic sets (NS), particle swarm optimization (PSO), and fast fuzzy C-mean (FFCM) algorithms. The method starts by improving image contrast using a median filter. The CT images are then transformed into the NS domain, evaluated by entropy, and processed using an optimized FFCM with PSO for segmentation. The segmented liver is further processed for tumor clustering and segmentation. Experimental results, using Jaccard Index, and Dice Coefficient metrics show the approach is accurate, time-efficient, noise-resistant, and performs well on non-uniform CT images.

3. Material and methods

CAD model using VGG based deep learning (VGGDL) is proposed. This approach involves intensity-based segmentation to effectively characterize the tumor lesion in CT liver images. In this paper, the proposed work is divided into two crucial phases: the initial stage involves evaluating various segmentation techniques like V-Net and U-Net. After that, the subsequent stage is dedicated to illustrating the

successful classification results, evaluating both qualitative and quantitative outcomes through the assessment of the VGG 16 model's performance.

3.1. Abdominal CT dataset: overview

In this study, a total of 564 CT images gathered from La Rabta Hospital of Tunis (218 BN Benign, 177 MLG malignant topics), including 169 depicting healthy conditions. The patient's median age was 54 years (between 35 and 80). We employed DICOM images sourced from 2D axial slices obtained from various CT scans. The use of standardized DICOM format ensures consistency in image quality and facilitates the integration of data, which is essential for accurate tumor identification and segmentation. The exported data are in JPG format with a size of 512*512 pixels. This study involved 250 patients with pathologically confirmed liver metastases and benign cases.

The data distribution for the hybrid V-Net-VGG16 model is depicted in Fig. 1.

A density filter using the (bone window) proves highly effective in examining liver structure and tumors, however, it does not permit working at a spatial resolution exceeding 2 mm in this dataset. Each patient presents a liver tumor with a measurement of 2 cm or less; considered in early stage. The tumor has not infiltrated the major blood vessels within the liver. A total of 564 CT images were employed for the training and validation of the proposed segmentation and classification model. The datasets containing liver lesions are segregated into two categories: 70 % allocated for training and 30 % for testing purposes. A substantial number of patients were used to directly train the proposed Encoder-Decoder CNN model [33] (see Fig. 2).

3.2. Data augmentation

Data augmentation is applied to the training dataset, emphasizing random transformations, including rotations, translations, scaling. New training instances were generated for every training epoch. Varied instances were created over 100 epochs to well train the deep neural network system. Indeed, solely linear transformations were utilized to prevent non-real cancers in the validation process.

3.3. Liver lesions segmentation phase

The liver cancer segmentation stage follows a dual-phase procedure, as outlined in Ref. [33]. It comprises initially an automated liver localization and secondly, the segmentation of the liver tumors regions. Because, tumor images depict tissues outside the liver with intensity analogous to that of cancers; its segmentation is so challenging task. The

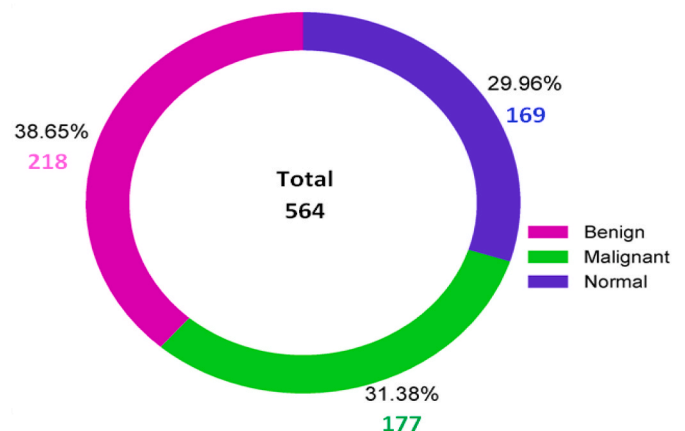


Fig. 1. Data distribution among different classes (Benign, Malignant, and Normal) represented in the dataset used for training and validation.

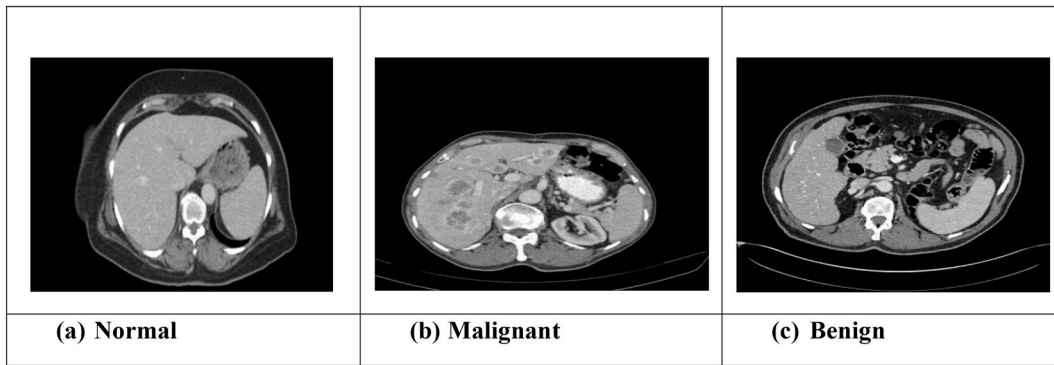


Fig. 2. An example of the training CT database representing different classes (Normal, Malignant and Benign cases).

initial input is streamlined using preceding location data and the probability intensity variation in the liver region through histogram equalization. A similar intensity to the liver region implies that the presence of other organs in the simplified image may interfere with the tumor segmentation process. Indeed, deep learning techniques can address the challenge of medical image processing.

It results in effectively avoiding classical segmentation approaches like mathematical morphologies and active contours. The central point of the most extensive connected area in the processed CT image is established as the initial point for region expansion.

3.3.1. V-net method

The V-Net method was deemed as an efficient CNN architecture for pixel-wise semantic segmentation. The applied approach is widely recognized in various medical scene understanding applications, including organs and tumor analysis, as investigated by the CNN layers in the VGG16 [36] architecture. The encoder layers within the V-Net architecture are used. Therefore, the decoder supplies max-pooling indices preserved and transmitted for the respective encoder layers.

V-Net is alternatively recognized as a volumetric ConvNet and was initially introduced in 2016 by Milletari et al. [37]. The architecture is inspired from the conventional 2D architecture of U-Net, but V-Net has the capability to handle 3D images. Additionally, with the assistance of residual layers, it better detects crucial features of the images (see Fig. 3).

Attention gates are often incorporated into V-net architectures to improve segmentation by focusing on significant features and regions in the image.

These gates operate in a spatial or channel-wise manner and are typically integrated into the skip connections of V-net, improving the model's ability to capture fine-grained details while focusing on significant features and regions in the image.

The Steps to integrate Attention Gate in V-Net method.

3.3.1.1. Attention gates (AG) in the skip connections. Attention gates are embedded into the skip connections between the encoder and decoder blocks.

Instead of directly passing feature maps from the encoder to the decoder, these gates refine the feature maps by focusing on relevant spatial or channel-wise information.

The attention gates are placed before the concatenation operations in the decoder:

Input to AG: feature map from the encoder (e.g $64 \times 64 \times n$).

Output: A refined feature map passed to concatenation step.

The skip connections are modified with attention gates. Instead of directly concatenating encoder feature maps with decoder outputs, the attention gate outputs weighted feature maps, emphasizing relevant parts before they are passed to the decoder.

3.3.2. U-net method

Providing swift and precise 2D-3D image segmentation [33], U-Net characterizes a fully CNN architecture.

The primary strength of this theory lies in its potency and efficient outcomes, even when applied to a limited training dataset. The U-Net architecture is divided into two stages: initial down-sampling (SAMPLING) and subsequent up-sampling (UP-SAMPLING).

The upper portion represents a typical fully CNN. It comprises a sequence of blocks with 3×3 convolution layers arranged as follows: An activation function employing rectified linear units (ReLU).

- A max pooling using a 2×2 filter and two strides for downsampling.

The number of features is multiplied by two after each block.

The feature maps are divided into two, after each block and linked with the corresponding feature map on the left side, as depicted by.

- A 3×3 convolution.
- An activation function (ReLU).

In the last layer, a one-by-one convolution is applied with a sigmoid activation function. This step is employed to link each 32-component feature vector to the target class.

In this study, we accomplished 26 convolutional layers, encompassing a total of 9,239,681 training parameters.

3.3.3. Training, test and validation process

Since no specific dataset was available for evaluating and validating the classification model, we adopted a cross-validation approach [38]. In this approach, the CT datasets and their corresponding ground truth liver masks were divided into different subsets. Each subset comprised 15 % of the CT data, randomly selected for validation purposes from the entire dataset. Both the testing and validation sets were equally utilized. We conducted training and testing for the classification method over a total of 13 folds. Subsequently, the mean accuracy was calculated by averaging the accuracy values obtained from the test datasets using the corresponding trained classification models.

3.4. An overview of the implementation classification method

The expected masks for abnormal liver tumors were generated by the classification model using the CT images.

These liver tumor masks were then used to identify characteristic CT images for three classes (Benign (BN) and malignant (MLG)) from the normal (NL) category. Patients were selected so that the disease's lesion areas, estimated from the corresponding predicted masks, comprised roughly 50 % of the largest cross-sectional area within the entire CT image for each patient.

This criterion eliminates false positive lesion predictions from the

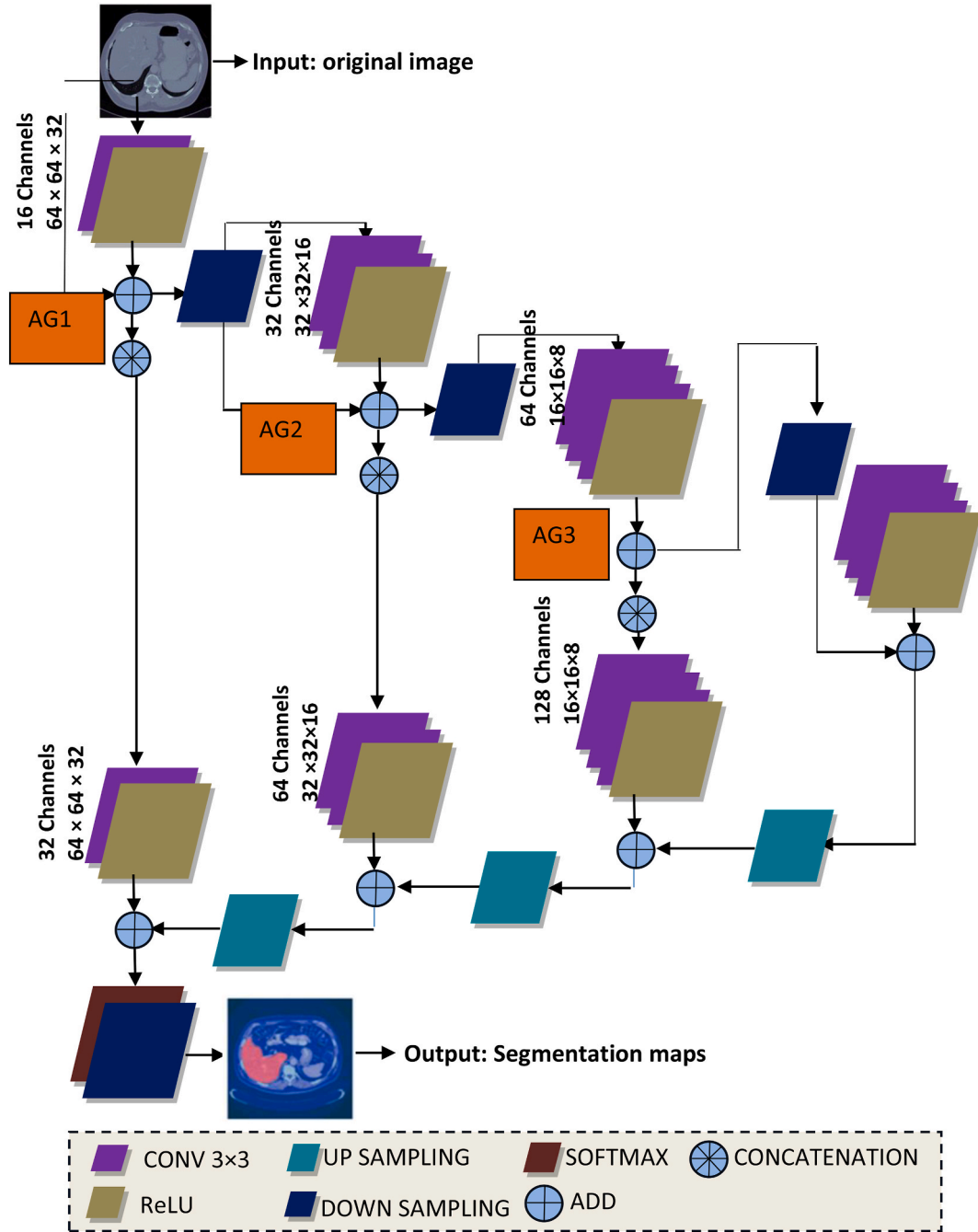


Fig. 3. V-Net architecture.

classification model, ensuring that the selected images contain relevant information about liver lesions, including details about area size and irregularities in infected ROI shape. This data information is typically concentrated near the center of the abnormal lesion and gradually reduces in importance towards the irregular boundaries.

The selected CT images were displayed with a Bone window to clearly depict the liver region with diseases and normal conditions (564 in total), each featuring an image size of $(256 \times 256 \times 3)$, where the images were divided into testing and training datasets and then employed for the training and validation of the classification method. The used optimizer was Adam, with a learning rate configured to 5×10^{-5} , employing binary accuracy representing by the model metric and binary cross-entropy by the loss function. After different experiences, the training process use 100 epochs with a batch size of 8. The uniform data augmentation was implemented, encompassing zoom, and shift

operations with a 40 % range, to moderate the risk of overfitting with a restricted CT data.

After the training process, the grading model was used to forecast the classes of liver tumors for the chosen CT images of each examined patient.

3.4.1. Classification model in liver tumors

As depicted in Fig. 4, the grading approach utilized VGG-16, with a classifier pre-trained using a convolutional system based on the VGG-16 network. The convolutional weights for the VGG-16 network can be obtained from the research conducted in Ref. [39].

The convolutional base contains five groups of convolutional layers with ReLU activation functions, each succeeded by a max-pooling layer.

The suggested classification approach consists of a flatten layer to transform the data from 3D to 1D, and a Dropout layer to correct the

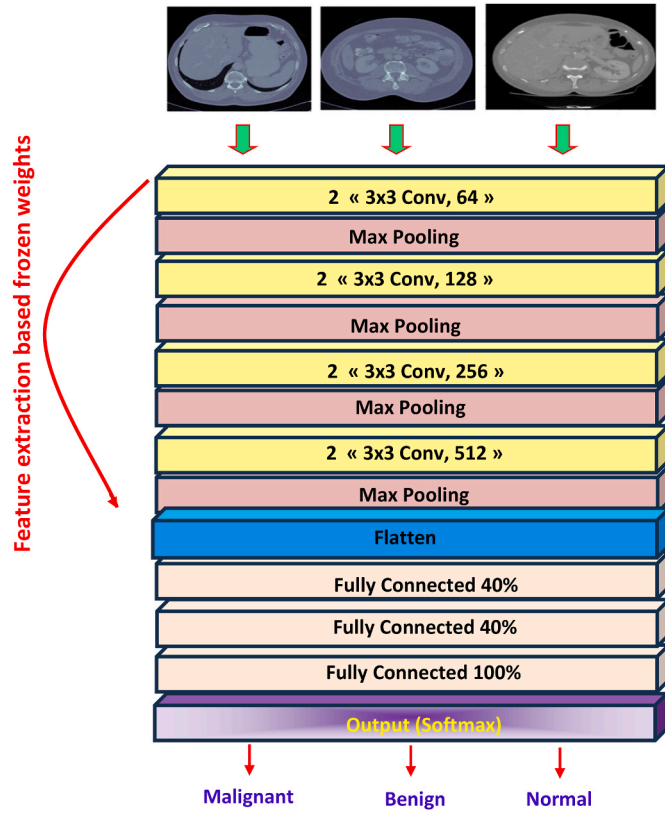


Fig. 4. The proposed methodology using VGG-16 for liver tumors classification.

problem of overfitting and eventually a connected layer using the Sigmoid activation function serving as the last layer (output).

Both of the input and output layers present dimensions of $256 \times 256 \times 3$ and 1, respectively the grading model was developed using Keras in combination with TensorFlow [40]. The employed datasets have been categorized into three classes. The data augmentation is implemented on the training CT images through random operations, including rotations and translations. Additional training instances were generated in each training epoch (Total of 100 epochs). In the validation system, linear transformations are applied to avoid the identification of non-real liver regions.

3.4.2. The proposed liver tumors segmentation and classification using V-NET-VGG-16 model

As shown in Fig. 5, the classification permits the categorization of the subject into three groups based on different liver tumors: BN, MLG and normal.

Using the VGG-16 classifier, the classification process is separated into two distinct phases: (i) the training stage, where the majority of the database (70 %) is applied to train the network.

However, in the testing and validation phase (ii), the VGG-16 technique is applied to the smaller portion of the CT image database (30 %). It is obvious that the training set surpasses the test set in size, a deliberate choice to ensure a meaningful learning process for the network and achieve enhanced classification results. During the training iterations, the architecture of VGG16 is refined.

Our network architecture includes an input layer with a reduced number of features (2 features after processing by the VGG-16 model), a single hidden layer with 30 neurons, and an output layer [41]. The desired outputs are defined as '0.5, 0.5, 0.5' for the BN case, '0.5, -0.5, 0.5' for the MLG case, and '0.5, 0.5, -0.5' for the normal case.

To determine the optimal number of neurons in the hidden layer, we conducted a series of experiments aiming to minimize the error rate and maximize classification accuracy. The ideal configuration of the VGG-16 model is identified by selecting the size of the hidden layer that results in the lowest validation error.

• Convolutional Blocks and Pooling Layers in VGG-16

In the VGG-16 architecture, there are four convolutional blocks denoted as Conv-1 to Conv-4, depending on the architecture, the arrangement of specific layers within a convolutional block may vary. However, a convolutional block typically incorporates one or more 2D convolutional layers (pooling layer). Then, the process of Flattening (Flatten) is applied to convert the multi-dimensional output of the convolutional and pooling layers into a 1D vector.

Extra layers are sometimes added, but for simplicity, we will focus on these two types of layers. Also, we have explicitly selected the last layer of the network as output (Softmax).

The softmax function is employed in this layer for the outputs from the final fully connected layer (FC). It normalizes the raw output values from the network within the range [0,1], allowing us to interpret them as probability scores for all classes.

3.5. Performance analysis

Different statistical measures can be used to control and evaluate the correctness and the effectiveness of the analysis results. In this study, the CNN architecture is appreciated to get a hopeful and truthful detection

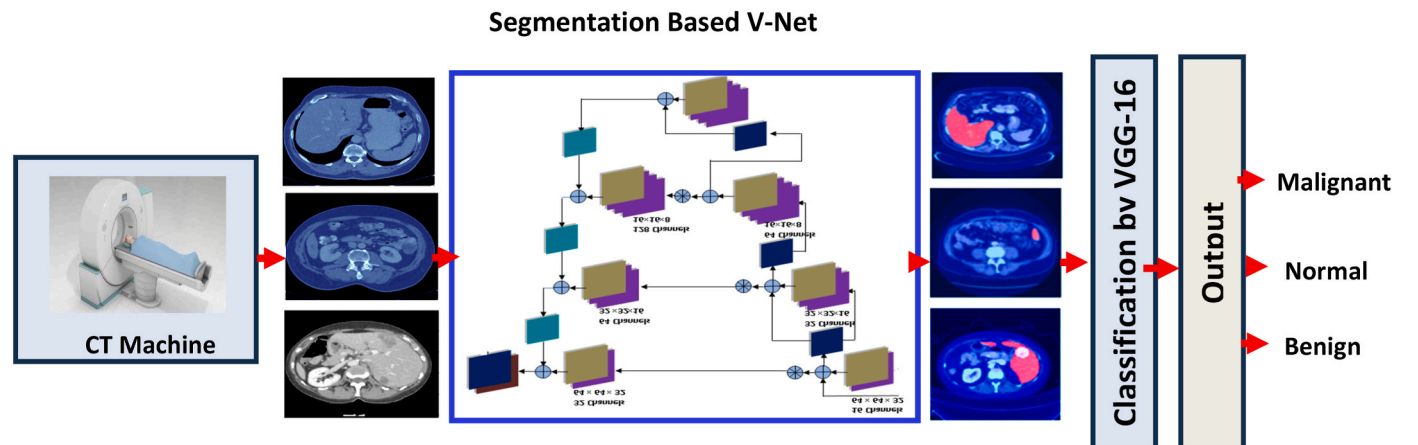


Fig. 5. The proposed hybrid (V-net-VGG16) approach.

ratio. The segmentations accuracy is computed focused on the resemblance degree between the ground truth and the resulting outputs projected by the two applied architectures (V-Net and U-Net). Hence, the included evaluation measure shown by the mean IoU is exposed by the following equation:

$$IoU = \frac{TP}{TP + FP + FN} \quad (1)$$

In our experiments, the evaluation of the proposed method is carried out using a statistical analysis in terms of accuracy 'AC', precision 'PC', recall and F1-score 'F1', of the liver tumor segmentation regions. These criteria are frequently used for the evaluation of analytical tests:

$$AC = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

$$PC = \frac{TP}{TP + FP} \quad (3)$$

$$recall = \frac{TP}{TP + FN} \quad (4)$$

$$F1 = \frac{2 \times PC \times recall}{PC + recall} \quad (5)$$

TP and TN: the true positive and true negative numbers, FP and FN: the false positive and false negative numbers, respectively.

Comparing actual performance in controlling tumor process, the one-sided Mann-Whitney *U* test is applied. This is a non-parametric test by verifying the null hypothesis. If the resulting segmentations and the ground truth derived from the same group, an important performance is achieved. In effect, a significant threshold is set to 0.05 which correspond to 95 % of confidence for accepting or not the null hypothesis.

4. Results of the segmentation and classification methods

In this study, we displayed an advanced experimentation to model hyper-parameter tuning based on two dissimilar structures of Encoder-Decoder architecture. A CNN (V-Net, U-Net) can strongly segment liver and tumors ROIs even if data augmentation placed in dissimilar orientations in order to have the property (invariance). Different experiments are realized to choose the optimum selected hyperparameters. In order to minimize the cost function, the stochastic gradient descent training method is applied to reduce the error rate between the actual and the desired output.

Tables 1 and 2 demonstrate the greatest significant parameters in terms of the obtained optimal values focused on the V-Net and U-Net.

Training process was proved the highest data (70 % of the database), the proposed techniques were validated on the remaining CT data (30 % of the remaining database) by computing the precision score (mean IoU).

In this study, we have made adjustments to the hyperparameters to optimize the segmentation of liver tumors. Specifically, we explored various configurations, including learning rate, batch size, number of layers, Training methods., to enhance model performance.

The mean IoU attained 96.74 % and 94.58 % using V-Net and U-Net, respectively. Besides, F1-score, Recall and precision are respectively 97.31 %, 96.84 % and 97.04 % for V-Net method compared to U-Net architecture 94.59 %, 93.83 % and 94.15 %. The resulting loss errors focused on binary cross entropy are 0.0024 and 0.036 for V-Net and U-Net, respectively. The segmentation results in term of qualitative

evaluation of V-Net and U-Net methods are exhibited in Fig. 6. In this context, we report that V-Net segmentation is closer to the ground truth compared to the obtained results of U-Net. We applied the statistical analysis to effectively demonstrate the relevant difference between the predicted segmented region and the ground truth labels supplied by the two models V-Net and U-Net. Simulation results illustrated in Fig. 6 highlight the robustness of the proposed V-Net method using large CT images. The projected approach was able to estimate tumor regions with an interesting accuracy result in the total database. Using V-Net model, experimental results are analyzed by computing the different statistical measures: the mean IoU, Recall 'Rec', precision 'PC' and F1_score 'F1'. From the obtained results, the performance of a segmentation algorithm usually depends on sensitivity and robustness; corresponding to the highest value of these statistical measure.

The p-values are computed between the ground truth and the two predicted segmentations. After several experiments, we can notice that is no important difference between the V-Net segmentation and the ground truth one which reaches a p-value of 0.031. Compared to V-Net architecture, a failed result is attained (p-value = 0.056) in order to determine the difference between the ground truth labels and the U-Net's segmentation. The two target segmentations furnished by V-Net and U-Net methods are different depending on *U* test. For the validation results, we applied a statistical analysis using Student's t-test in order to evaluate the difference between the predicted segmentation provided by the two architectures and the ground truth. In fact, if, p-value is inferior to 0.05 (p-value < 0.05) then, it was considered significant. Besides, for each experiment, the pertinent binary cross entropy is depended on the lower concerned value.

In this paper, we introduced an advanced approach based on two different fully CNN encoder-decoder architectures. As shown in Fig. 7, these V-Net and U-Net structures were effectively realized to accomplish liver tumors segmentation. It is clear that U-Net segmentation results are not highly near to the ground truth also it shows a divergent loss curves (see Fig. 8). In fact, the foremost variation between the two applied models is presented in their connection type in view the convolution and deconvolution. V-Net employed the kept pooling indices through convolution process compared to U-Net is focused on features maps linking in terms of convolution and deconvolution. In this work, the attendance of tumor lesion can be exposed in rapports of analysis reliability of the liver lesions localization. Three radiologists, with varied expertise degrees, contributed in specifying the liver tumor ground truth, via the attentive lesion detection on each patient.

Furthermore, we validated the proposed V-Net algorithm on an additional publicly available CT liver abdominal dataset to further assess its robustness [34,35]. The segmentation results on CT images of liver and lesions demonstrated the effectiveness of our method. The algorithm showed high accuracy in detecting and segmenting the tumors, confirming its potential for clinical applications. This validation highlights the adaptability of V-Net to different datasets and its ability to generalize well in diverse medical imaging contexts (see Fig. 9).

5. Discussion

The majority of current approaches used segmentation improvement techniques based on statistical form and emergence models.

This study of Sahli et al. [33] introduces a predictive segmentation method for detecting liver tumors in CT images, addressing challenges in metastasis lesion localization for diagnosis and treatment. Using an encoder-decoder structure, the approach enables volumetric analysis of

Table 1
Optimum results according the used hyper-parameters via V-Net and U-Net models.

Method	Data augmentation	Learning rate	Momentum rate	Batch size	Weight initialization	Training methods	Learning rate decay	Dropout rates
V-Net	Linear	0.0005	0.99	8	Normal	SGD	Linear (0.000062)	0.30
U-Net [24]	Linear	0.0005	0.99	8	Normal	SGD	Linear (0.000062)	0.10

Table 2
Tested hyper-parameters.

Hyper-parameters	Data		Learning rate					Batch size					Momentum rate			Weight initialization			Training methods			
Tests	L	nL	0.5	0.05	0.005	0.0005		2	4	8	16	32	0.8	0.9	0.99	N	U	GU	RMS	A	Ag	SGD

*Linear “L”, No-linear “nL”, Normal “N”, uniform “U”, Glorot uniform “GU”, RMSprop “RMS”, Adam “A”, Adagrad “Ag”, Stochastic gradient descent “SGD”.

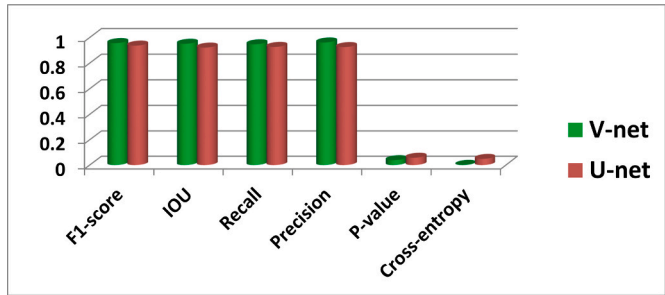


Fig. 6. The obtained statistical results in view of V-net and U-net segmentation architectures.

tumors, leveraging Seg-Net and U-Net architectures. A dataset of 200 confirmed metastasis cases, comprising 8297 CT slices, was used for model training (85 %) and validation (15 %). The method achieved high performance metrics: F1-score (0.9573), Recall (0.9520), IoU (0.9654), and Binary Cross-Entropy (0.0032), with a statistically significant p-value (<0.05). Compared to existing techniques, the proposed method demonstrates superior precision in identifying metastasis tumor locations.

The study of Khoshkhabar et al., introduces a novel deep learning-based technique for liver and tumor segmentation using CT images, leveraging the LiTS17 dataset. The model incorporates four Chebyshev graph convolution layers and a fully connected layer, achieving high performance metrics: Dice coefficient of 90.00 %, mean IoU of 89.2 %, and precision of 88.7 %.

Indeed, an advanced approach for liver lesions detection is applied in the work of Albishri et al. [43] based on multimodal U-net model. The

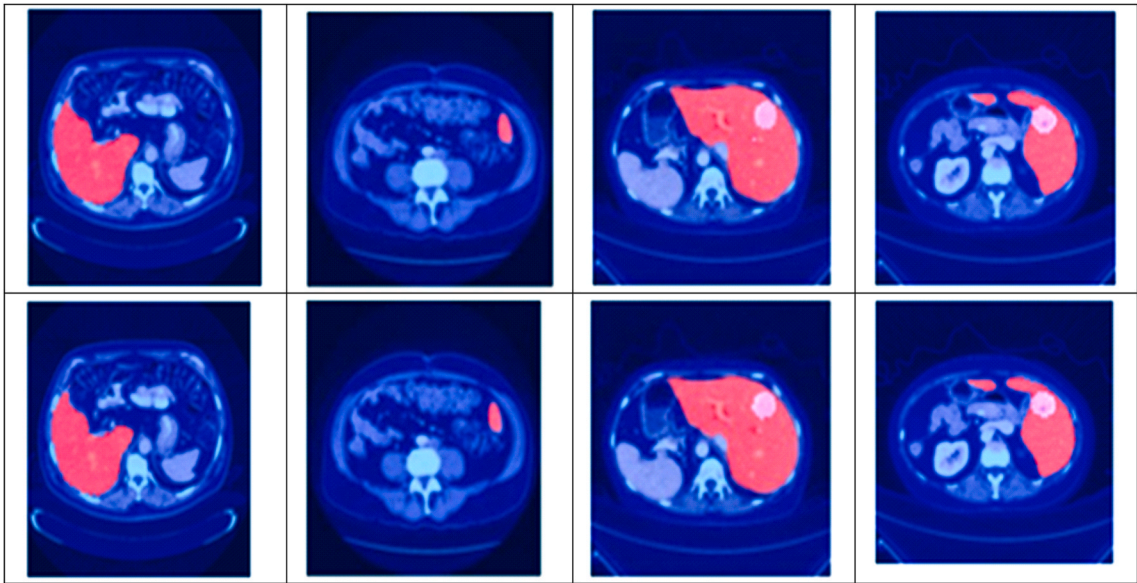


Fig. 7. Quantitative segmentation results using V-net (top row) and U-net (bottom row) architectures.

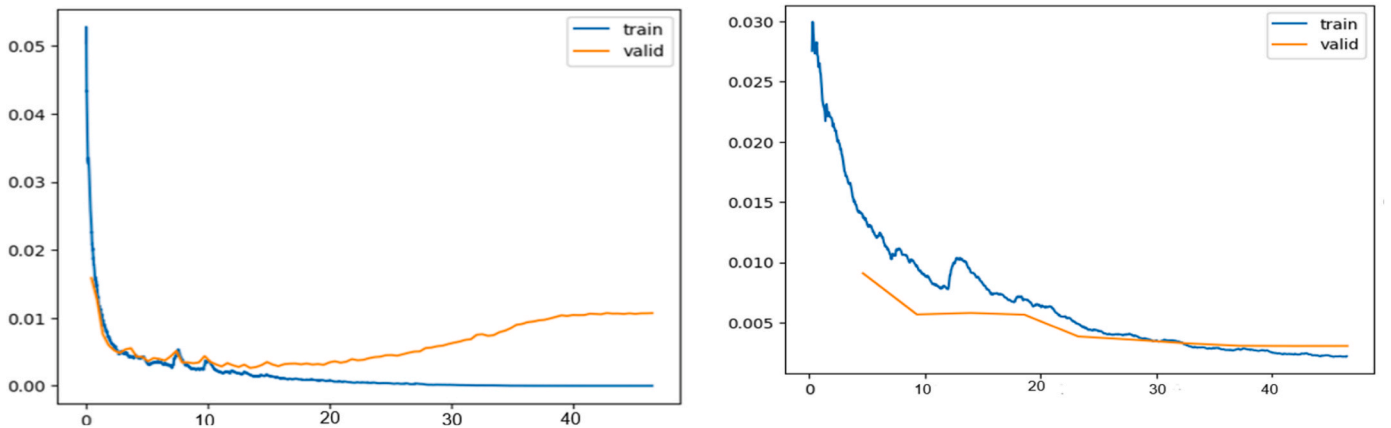


Fig. 8. Performance of the proposed V-Net (right) and U-net (left) training and validation Loss.

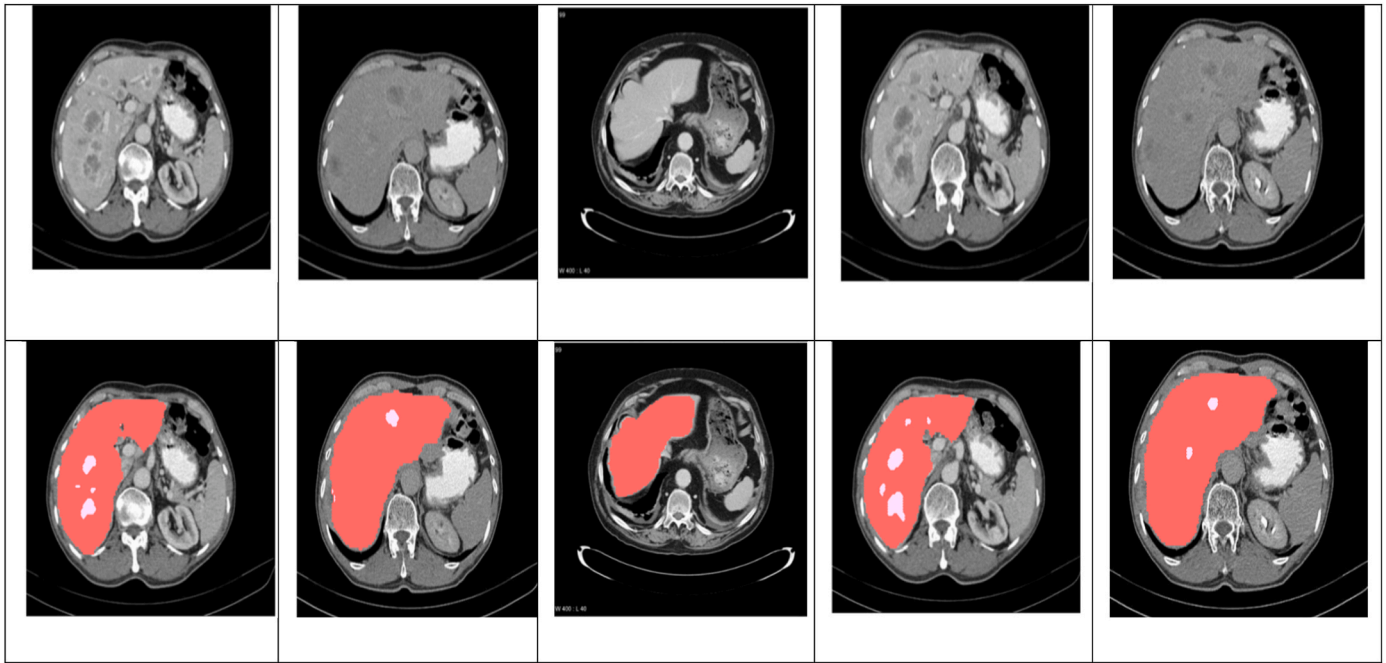


Fig. 9. Validation of the proposed V-net method on axial slices for liver CT images.

authors notice that the segmentation results are efficiency improved than those obtained by atlas-based methods. Still, deep learning methods confirm the high potential in the medical segmentation applications. But the learned features require to be trained on a greater data base. For an improved qualitative analysis, a statistical measurement could be used to compute the quality of image segmentation.

Besides, Hu et al. [48], propose a novel label-free tumor synthesis approach for unsupervised liver tumor segmentation, aiming to address the limitations of existing methods. By focusing only on the liver region during training and testing, the approach minimizes noise from extraneous information, improving segmentation performance. In contrast, the proposed method achieves a Dice Similarity Coefficient (DSC) of 59.77 %.

The work of Chen et al. [49], presents MS-FANet, a Multi-Scale Feature Attention Network designed to enhance liver tumor segmentation in CT images by addressing the challenge of varying tumor sizes. Key innovations include the Residual Attention (RA) Block and Multi-Scale Atrous Downsampling (MAD) for extracting multi-scale tumor features, as well as the Dual-Path Feature (DF) Filter and Dense Upsampling (DU) for optimizing feature reduction and improving segmentation accuracy. Tested on the LiTS and 3DIRCADb datasets, MS-FANet achieved average Dice scores of 74.2 % and 78.0 %, respectively, outperforming most state-of-the-art methods and demonstrating excellent adaptability and segmentation performance.

In other work of Li et al. [50], authors addressed the clinical need for automated liver tumor segmentation in CT images, where small tumors with unclear borders are often missed. An improved U-Net-based method is proposed, incorporating attention mechanisms in the skip connections to emphasize important regions dynamically. This enhancement enables precise localization and differentiation of target areas. Testing on the LiTS dataset achieved a Dice similarity coefficient of 69.00 %, recall of 63.5 %, and precision of 84.7 %, demonstrating the method's effectiveness in accurately segmenting liver tumors.

Table 3 reveals that the segmentation results of the proposed method demonstrate effective metric values using our dataset, with a Dice average of 97.34 %, IoU of 96.74 %, F1-score of 97.31 %, recall of 96.84 %, and precision of 97.04 %, compared to other tested state-of-the-art techniques [33,42,43], and [48–50].

Finally, the Hybrid segmentation and classification methodology can

Table 3

Summary of some studies reporting some liver lesion segmentation approaches using CT datasets.

Method	Dice (%)	IoU (%)	F1-Score (%)	Recall (%)	Precision (%)
V-Net (Proposed)	97.34	96.74	97.31	96.84	97.04
U-Net (Sahli et al., 2022) [33]	93.54	96.54	95.73	95.20	96.52
SLIC-DGN (Khoshkhabar et al., 2023) [42]	90.00	89.20	N/A	N/A	88.70
Cascaded U-Net (Albishi et al., 2019) [43]	89.40	N/A	N/A	N/A	N/A
U-Net (Hu et al., 2023) [48]	59.77	N/A	N/A	N/A	N/A
MS-FANet (Chen et al., 2023) [49]	74.20 78.00	N/A	N/A	N/A	N/A
U-Net (Li et al., 2023) [50]	69.20	N/A	N/A	63.50	84.70

**SLIC-DGN: SLIC-Based Deep Graph Network.

**MS-FANet: Multi-Scale Feature Attention Network.

faithfully be exploited for the analysis of liver tumors in precocious stage and used consequently as a further tool to assist radiologist in the diagnosis process.

Fig. 10 depict the mean classification accuracy, sensitivity and precision of the proposed CNN models featuring convolution layers, using a configuration of 9 hidden layers. It's noteworthy that the increase in the number of convolution layers, ranging from 3 to 9 hidden layers, contributes to the improvement in model precision. Consequently, for all the models used in this study (Dense-Net, Res-Net and VGG-16), data augmentation has been implemented.

A confusion matrix is used to summarize the diagnostic precision of the used VGG16 model (see Fig. 11).

To identify liver cancers using CT images, diverse methods are applied to the testing set in terms of the testing process [44,45]. Among these, the proposed VGG-16-based approach is evaluated using various metrics, including accuracy, sensitivity, specificity, and precision. The

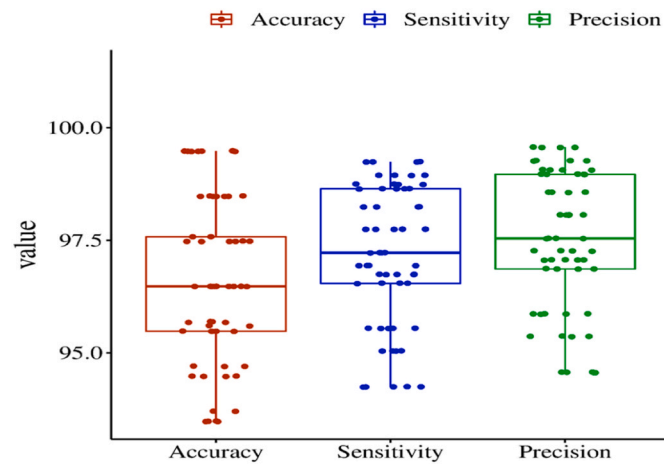


Fig. 10. Results of the computer-assisted VGG-16 liver cancer on the complete CT dataset (mean values $\pm$ standard deviation).

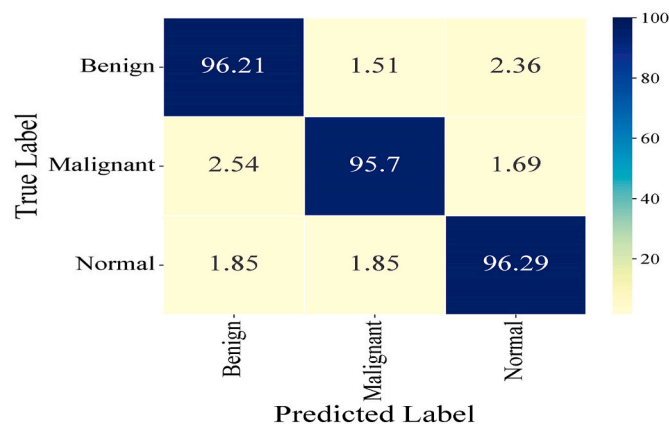


Fig. 11. Confusion matrix for the prediction models presenting the precision of the true and predicted liver tumors from CT images data.

evaluation includes three classes (BN, MLG, and normal) by computing these metrics for each patient. The proposed method is then compared to other computer-aided methods such as Res-Net [46] and Dense-Net [47]. All metrics are calculated for each patient in the dataset, and the average values across all patients are determined. As depicted in Fig. 11, the proposed method demonstrates remarkable model accuracy in terms of both accuracy and loss measures, achieving the highest accuracy results.

The results of liver tumors classification are described in Fig. 12. We display the efficiency of the proposed method established on the VGG-16 procedure employing a large CT dataset. The proposed system was able to distinguish liver tumor types (BN, MLG compared to Normal) with a precision average of 97.49 %, F1score of 97.16 % and recall of 96.75 % for VGG-16 in the test CT dataset. From these results, we conclude that the proposed VGG-16 classification technique can reliably be employed to help neurologist doctors by supplying a second analysis strategy for unknown cases that require additional attention.

As depicted in Fig. 12, for the categorization of liver tumor classes using CT images, we employed the identified texture regions in the CT images. The categorization of distinctive liver tumors is implemented utilizing VGG-16. These results are comparable to the liver tumor classes discrimination (BN, MLG) and normal cases obtained by three suggested classification models from the literature using Res-Net [46], Dense-Net [47]. The models were assessed without any modifications to their parameters and hyperparameters.

The proposed VGG16 model used 564 images for the classification

model, employing a cross-validation approach with 13 folds. In this approach, 9 folds were allocated for training and 4 for testing. The selection of 1 out of 10 cross-validation sets strikes a balance between the number of train-test datasets and the size of the training data required for accurately estimating model accuracy [50,51].

To further justify this approach, we have conducted additional experiments where the classification task was performed directly on unsegmented CT images. The results showed a noticeable difference in accuracy compared to the hybrid V-Net-VGG16 model. Specifically, the classification accuracy decreased from 96.52 % (with segmentation process) to 89.34 % (without segmentation process). These findings confirm that segmentation contributes substantially to the accuracy and robustness of the proposed analysis by providing cleaner, more focused inputs for the classification technique [52].

At the level of CT images, the performance of the proposed classification model is comparable to that reported in Refs. [46,47], even though the classification involved two liver diseases. Notably, it surpasses the reported accuracy value of 95.00 in reference [46], which used CT image datasets. The presented models bring forth certain concerns that could be addressed in future enhancements. The primary objective is to enhance the generalization of the models, particularly the discrimination model, given the relatively limited number of images. Therefore, it would be valuable to investigate and compare the results from these network techniques with the proposed architecture. Additionally, the proposed VGG16 learning used a relatively straightforward network architecture based on a hybrid design, which is deemed suitable for the dataset used in the current work for training and validation.

It will be our future research to extend this approach to not only classifies CT images into 3 categories but also to detect the disease's features within images. Many other tumors region can be investigated (e.g., breast cancer, lung cancer. In the following, we will show that our proposed strategy and the model can classify CT images with minimal error.

6. Conclusion

In this paper, we proposed automated liver tumor segmentation and classification approaches based on two different encoder-decoder architectures: V-Net and VGG16. In each architecture, all layers and parameters are rectified in order to repeat the deep neural network training based on computed tomography data. To enhance the forcefulness of the proposed model, a novel data can be added that is still following annotation. We plan to use the segmented liver cancer approach for advanced tumor stages to efficiency improve prediction system and provide a prospective validation. Besides, the proposed segmentation method of liver tumors regions can be deployed to other lesion organs such as breast cancer response.

CRedit authorship contribution statement

Amine Ben Slama: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Hanene Sahli:** Writing – review & editing, Visualization, Validation, Investigation. **Yessine Amri:** Writing – review & editing, Visualization, Methodology, Investigation. **Salam Labidi:** Writing – review & editing, Supervision, Project administration.

Ethics approval and consent to participate

This study was conducted in accordance with the principles of the Declaration of Helsinki established by the World Medical Association and approved by the Human Ethics Committee of Bechir Hamza Children's Hospital of Tunis.

All methods were conducted in compliance with relevant guidelines and regulations.

Informed consent to participate was obtained from all participants

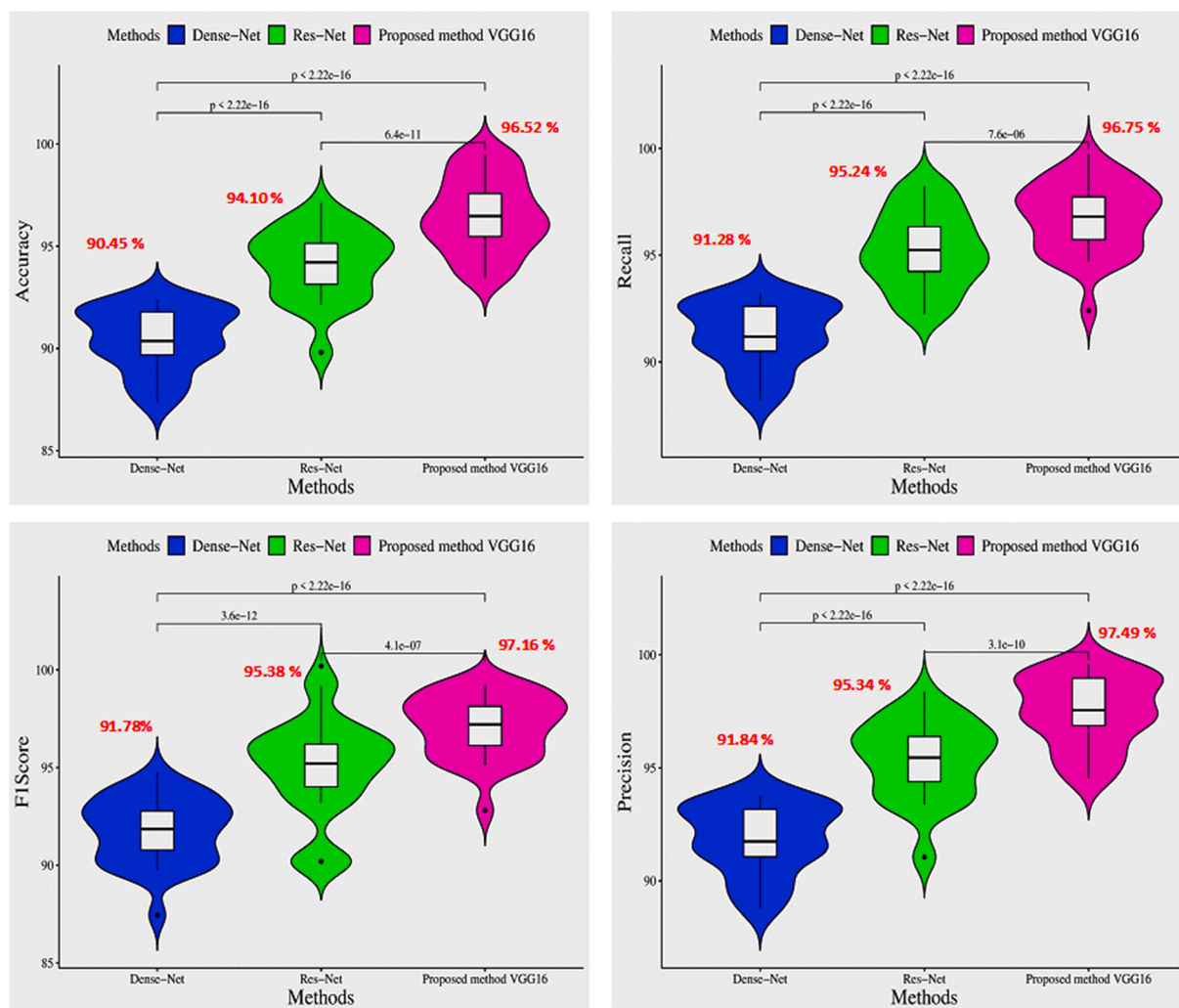


Fig. 12. Violin plots of the accuracy, sensitivity, recall, F1-score, precision distribution for the CT data sets used for validation of classification model using Dense-Net Res-Net50 and the proposed VGG16.

after receiving all the necessary information about the research. Participants consented to have their X ray images published. All data and identities of patients were processed with strict confidentiality.

Consent for publication

Written informed consent for publication of their clinical details and/or clinical images was obtained from the parents.

Availability of data and materials

For Data evaluation please contact Dr. Ben Slama Amine at; amine.ben.slama@istmt.utm.tn.

All data analyzed during this study are available upon request.

Ethics statement

Ethics approval and consent to participate: This study was conducted in accordance with the principles of the Declaration of Helsinki established by the World Medical Association and approved by the Human Ethics Committee of Bechir Hamza Children's Hospital of Tunis (reference number: HET11/24).

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] Kulkarni NM, Fung A, Kambadakone AR, Yeh BM. Computed tomography techniques, protocols, advancements, and future directions in liver diseases. *Magnetic Resonance Imaging Clinics* 2021;29(3):305–20.
- [2] Ananth C, Roshni Bai D, Renuka K, Vidhya A, Savithra C. Liver and hepatic tumors segmentation in 3-D CT images. *International Journal of Advanced Research in Computer Engineering & Technology (IJARCET)* 2014;3(2).
- [3] Ho JL, Konda A, Rahman J, Harris E, Korn R, Sabir A, Bawany B, Gulati R, Harris GJ, Boswell WD, Fong Y. Comparative analysis of three-dimensional volume rendering and maximum intensity projection for preoperative planning in liver cancer. *European Journal of Radiology Open* 2020;7:100259. <https://doi.org/10.1016/j.ejro.2020.100259>.
- [4] Mise Y, Tani K, Aoki T, Sakamoto Y, Hasegawa K, Sugawara Y, Kokudo N. Virtual liver resection: computer-assisted operation planning using a three-dimensional liver representation. *J Hepato-Biliary-Pancreatic Sci* 2013;20:157–64.
- [5] Zhou T, Ruan S, Canu S. A review: deep learning for medical image segmentation using multi-modality fusion. *Array* 2019;3:100004.
- [6] Chen Y, Yue X, Fujita H, Fu S. Three-way decision support for diagnosis on focal liver lesions. *Knowl Base Syst* 2017;127:85–99.
- [7] Falk T, Mai D, Bensch R, Çiçek Ö, Abdulkadir A, Marrakchi Y, Ronneberger O. U-Net: deep learning for cell counting, detection, and morphometry. *Nat Methods* 2019;16(1):67–70.
- [8] Huang Q, Luo Y, Zhang Q. Breast ultrasound image segmentation: a survey. *Int J Comput Assist Radiol Surg* 2017;12:493–507.
- [9] Lebre MA, Vacavant A, Grand-Brochier M, Rositi H, Abergel A, Chabrot P, Magnin B. Automatic segmentation methods for liver and hepatic vessels from CT and MRI volumes, applied to the Couinaud scheme. *Comput Biol Med* 2019;110:42–51.
- [10] Chen L, Song H, Wang C, Cui Y, Yang J, Hu X, Zhang L. Liver tumor segmentation in CT volumes using an adversarial densely connected network. *BMC Bioinf* 2019;20:1–13.
- [11] Ahn JC, Connell A, Simonetto DA, Hughes C, Shah VH. Application of artificial intelligence for the diagnosis and treatment of liver diseases. *Hepatology* 2021;73(6):2546–63.
- [12] Huang Q, Ding H, Wang X, Wang G. Fully automatic liver segmentation in CT images using modified graph cuts and feature detection. *Comput Biol Med* 2018;95:198–208.
- [13] Kumar SS, Moni RS, Rajeev J. Liver tumor diagnosis by gray level and contourlet coefficients texture analysis. In: 2012 international conference on computing, electronics and electrical technologies (ICCEET); 2012, March. p. 557–62. IEEE.
- [14] Priya V, Umamaheswari K. Enhanced continuous and discrete multi objective particle swarm optimization for text summarization. *Cluster Comput* 2019;22:229–40.
- [15] Cai L, Gao J, Zhao D. A review of the application of deep learning in medical image classification and segmentation. *Ann Transl Med* 2020;8(11).
- [16] Yasaka K, Akai H, Abe O, Kiryu S. Deep learning with convolutional neural network for differentiation of liver masses at dynamic contrast-enhanced CT: a preliminary study. *Radiology* 2018;286(3):887–96.
- [17] Zhao G, Li G, Xu R, Lin L. Collaborative training between region proposal localization and classification for domain adaptive object detection. In: *Computer vision—ECCV 2020: 16th European conference, glasgow, UK, august 23–28, 2020, proceedings*, Part XVIII 16. Springer International Publishing; 2020. p. 86–102.
- [18] Yang R, Yu Y. Artificial convolutional neural network in object detection and semantic segmentation for medical imaging analysis. *Front Oncol* 2021;11:638182.
- [19] Sahli H, Mouelhi A, Ben Slama A, Sayadi M, Rachdi R. Supervised classification approach of biometric measures for automatic fetal defect screening in head ultrasound images. *J Med Eng Technol* 2019;43(5):279–86.
- [20] Algarni AD. Automated medical diagnosis system based on multi-modality image fusion and deep learning. *Wireless Pers Commun* 2020;111(2):1033–58.
- [21] Melendez J, Van Ginneken B, Maduskar P, Philipsen RH, Reither K, Breuninger M, Sánchez CI. A novel multiple-instance learning-based approach to computer-aided detection of tuberculosis on chest X-rays. *IEEE Trans Med Imag* 2014;34(1):179–92.
- [22] Huo Y, Terry JG, Wang J, Nair S, Lasko TA, Freedman BI, Landman BA. Fully automatic liver attenuation estimation combining CNN segmentation and morphological operations. *Med Phys* 2019;46(8):3508–19.
- [23] Esteve A, Kuprel B, Novoa RA, Ko J, Swetter SM, Blau HM, Thrun S. Dermatologist-level classification of skin cancer with deep neural networks. *Nature* 2017;542(7639):115–8.
- [24] Wu B, Fang Y, Lai X. Left ventricle automatic segmentation in cardiac MRI using a combined CNN and U-net approach. *Comput Med Imag Graph* 2020;82:101719.
- [25] Tang W, Zou D, Yang S, Shi J, Dan J, Song G. A two-stage approach for automatic liver segmentation with Faster R-CNN and DeepLab. *Neural Comput Appl* 2020:1–10.
- [26] Elshaer MEA, Wisdom S, Mishra T. Transfer learning from sound representations for anger detection in speech. 2019. *arXiv preprint arXiv:1902.02120*.
- [27] Sharma A, Liu X, Yang X, Shi D. A patch-based convolutional neural network for remote sensing image classification. *Neural Network* 2017;95:19–28.
- [28] Moghbel M, Mashohor S, Mahmud R, Saripan MIB. Automatic liver segmentation on computed tomography using random walkers for treatment planning. *EXCLI journal* 2016;15:500.
- [29] Kavur Ali Emre, Alper Selver M, Dicle Oğuz, Barış Mustafa, Sinem Gezer N. CHAOS - Combined (CT-MR) Healthy Abdominal Organ Segmentation Challenge Data (Version v1.03) [Data set]. Zenodo. 2019. <http://doi.org/10.5281/zenodo.3362844>.
- [30] Atika L, Nurmaini S, Partan RU, Sukandi E. Image segmentation for mitral regurgitation with convolutional neural network based on unet, resnet, vnet, fractalnet and segnet: a preliminary study. *Big Data and Cognitive Computing* 2022;6(4):141.
- [31] Kushnure DT, Talbar SN. MS-UNet: a multi-scale UNet with feature recalibration approach for automatic liver and tumor segmentation in CT images. *Comput Med Imag Graph* 2021;89:101885.
- [32] Bilic P, Christ P, Li HB, Vorontsov E, Ben-Cohen A, Kaissis G, Menze B. The liver tumor segmentation benchmark (lits). *Med Image Anal* 2023;84:102680.
- [33] Sahli H, Ben Slama A, Labidi S. U-Net: a valuable encoder-decoder architecture for liver tumors segmentation in CT images. *J X Ray Sci Technol* 2022;30(1):45–56.
- [34] Anter AM, Bhattacharyya S, Zhang Z. Multi-stage fuzzy swarm intelligence for automatic hepatic lesion segmentation from CT scans. *Appl Soft Comput* 2020;96:106677.
- [35] Anter AM, Hassenian AE. Computational intelligence optimization approach based on particle swarm optimizer and neutrosophic set for abdominal CT liver tumor segmentation. *Journal of Computational Science* 2018;25:376–87.
- [36] Younis A, Qiang L, Nyatega CO, Adamu MJ, Kawuwa HB. Brain tumor analysis using deep learning and VGG-16 ensembling learning approaches. *Appl Sci* 2022;12(14):7282.
- [37] Milletari F, Navab N, Ahmadi SA. V-net: fully convolutional neural networks for volumetric medical image segmentation. In: *2016 fourth international conference on 3D vision (3DV)*. IEEE; 2016, October. p. 565–71.
- [38] Slama AB, Sahli H, Amri Y, Trabelsi H. Res-Net-VGG19: improved tumor segmentation using MR images based on Res-Net architecture and efficient VGG gliomas grading. *Applications in Engineering Science* 2023;16:100153.
- [39] Siddique N, Paheding S, Elkin CP, Devabhaktuni V. U-net and its variants for medical image segmentation: a review of theory and applications. *IEEE Access* 2021;9:82031–57.
- [40] Panja A, Christy JJ, Abdul QM. An approach to skin cancer detection using Keras and Tensorflow. In: *Journal of physics: conference series*. IOP Publishing; 2021, May, 012032. Vol. 1911, No. 1.
- [41] Sahli H, Ben Slama A, Zerai A, Labidi S, Sayadi M. ResNet-SVM: fusion based glioblastoma tumor segmentation and classification. *J X Ray Sci Technol* 2023;31(1):27–48.
- [42] Khoshkhabar M, Meshgini S, Afrouzian R, Danishvar S. Automatic liver tumor segmentation from CT images using graph convolutional network. *Sensors* 2023;23(17):7561.
- [43] Albishri AA, Shah SJH, Lee Y. CU-Net: cascaded U-Net model for automated liver and lesion segmentation and summarization. In: *2019 IEEE international conference on bioinformatics and biomedicine (BIBM)*. IEEE; 2019, November. p. 1416–23.
- [44] Anter AM, Abualigah L. Deep federated machine learning-based optimization methods for liver tumor diagnosis: a review. *Arch Comput Methods Eng* 2023;30(5):3359–78.
- [45] Thakare A, Pillai S, Neman R, Shitkurkar N, Nair A, Anter AM. A system for liver tumor detection. In: *International conference on information, communication and computing technology*. Cham: Springer Nature Switzerland; 2023, May. p. 129–41.
- [46] Showkat S, Qureshi S. Efficacy of transfer learning-based ResNet models in chest X-ray image classification for detecting COVID-19 pneumonia. *Chemometr Intell Lab Syst* 2022;224:104534.
- [47] Zhou T, Ye X, Lu H, Zheng X, Qiu S, Liu Y. Dense convolutional network and its application in medical image analysis. *BioMed Res Int* 2022;2022.
- [48] Hu Q, Chen Y, Xiao J, Sun S, Chen J, Yuille AL, Zhou Z. Label-free liver tumor segmentation. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*; 2023. p. 7422–32.
- [49] Chen Y, Zheng C, Zhang W, Lin H, Chen W, Zhang G, Wu F. MS-FANet: multi-scale feature attention network for liver tumor segmentation. *Comput Biol Med* 2023;163:107208.
- [50] Li H, Liang B. Liver tumor computed tomography image segmentation based on an improved U-Net model. *Appl Sci* 2023;13(20):11283.
- [51] Jin Q, Cui H, Sun C, et al. Free-form tumor synthesis in computed tomography images via richer generative adversarial network. *Knowl Base Syst* 2021;218:106753.
- [52] Jin Q, Cui H, Sun C, et al. Domain adaptation based self-correction model for COVID-19 infection segmentation in CT images. *Expert Syst Appl* 2021;176:114848.